

Does BETER's Odds Feed Lead Kalshi?

Option C — BETER-Internal Microstructure

Interim results: Parts 0–2 (Part 3 edge backtest forthcoming)

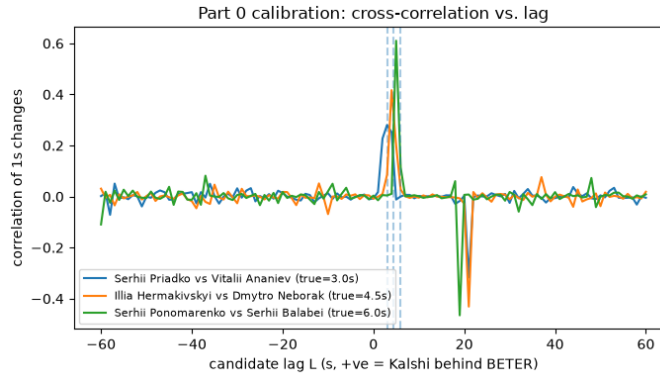
Scope

Capture spans 2026-08-28/29: ~750 Setka Cup table-tennis (TT) fixtures, ~2,000 eFootball (FIFA) fixtures, and 3 ESEA Counter-Strike matches on BETER, alongside 276 open esports markets on Kalshi. **Kalshi listed no TT series during this capture window** (confirmed in Part 1's crosswalk: 0 of 543 TT matches found a live Kalshi counterpart), which is the assignment's sanctioned trigger to fall back to **Option C**: characterize BETER's own internal microstructure — incident→trading latency, price-jump size by incident type and score state, and calibration of BETER's margin-free probability against realized settlement — rather than a cross-venue lead/lag that this dataset cannot support for TT.

Part 0: Sanity Harness

Before trusting any lag estimate on real data, the estimator itself is validated against `demo/make_synthetic.py`'s synthetic capture, which has known, injected lags (3.0/4.5/6.0s).

Match	True lag (s)	Estimated lag (s)	Recovery error (s)	Peak corr.
Priadko vs Ananiev	3.0	3.0	0.0	0.280
Hermakivskyi vs Neborak	4.5	4.0	−0.5	0.417
Ponomarenko vs Balabei	6.0	5.0	−1.0	0.610



Resolution floor: ± 1 second. The estimator (1s grid, step/fill interpolation, cross-correlate 1s changes over ± 60 s) recovers 3.0s exactly and 4.5s/6.0s to within 1s — a 1Hz grid can only return an integer lag, so a non-integer true lag necessarily rounds to the nearest grid point. Any lag reported from real data should be read as ± 1 s. (Peak correlations are of a synthetic, deliberately noisy signal, not a real-data benchmark — see appendix A.)

Part 1: Wrangling

Manifest: 2,818 booked fixtures (2,060 Fifa, 755 TT, 3 CS). Of these, 1,465 produced at least one **trading** or **incident** message in the capture window (919 Fifa, 543 TT, 3 CS) and form the per-match deliverable table.

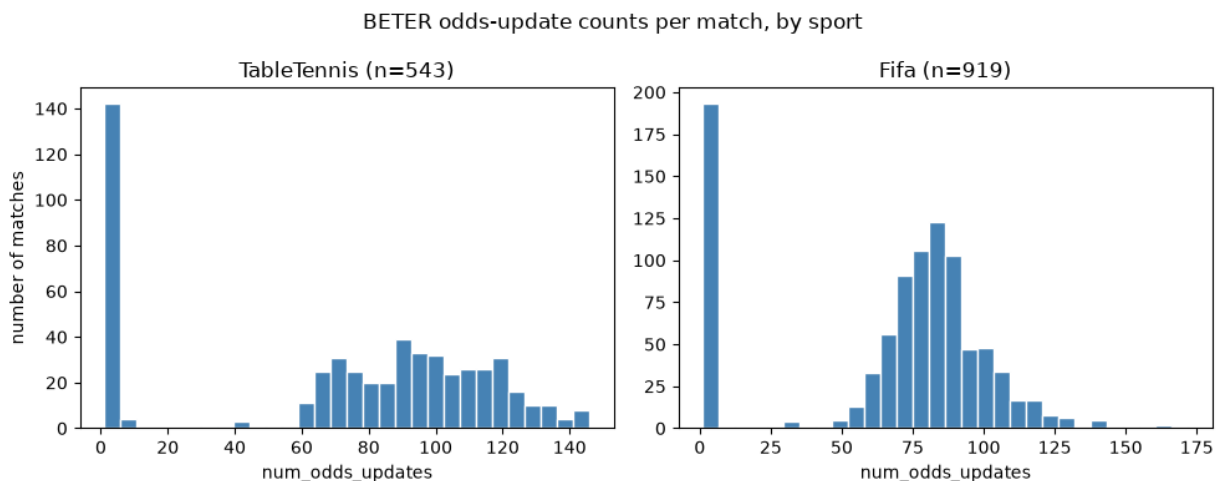
Data pitfalls identified and handled:

- **Two incident-index namespaces share one field name.** A handful of administrative types (2,3,5,7,17,18) use a tiny, separate local counter while the real per-match scoring sequence is a large shared per-sport counter;

pooling both blindly makes the gap analysis show 99.9% "missing." Fix: exclude admin types first. TT is then ~1% missing (credible); FIFA's ~57% "missing" is *not* real loss — our booked fixtures are a small slice of a much larger global eFootball stream, so most "missing" values belong to matches we never booked. Only the TT number is trusted as a loss estimate.

- **CounterStrike: the incident channel delivered zero point-level events** across all 3 CS matches — a real coverage finding, not a bug.
- **Kalshi price fields** appear only as *.dollars strings in this capture; the cents-int fields never appear. Reading only the int fields silently returns `None`, which looks exactly like "every book is empty" but is a parsing bug, not a market finding.
- **Kalshi ticker timestamps are America/New York local time**, not UTC — off by ~4h until corrected, which broke the BETER↔Kalshi crosswalk until diagnosed.
- **Only 2 of 1,465 BETER matches ever had a live, simultaneous Kalshi market** (both Counter-Strike) — the concrete evidence behind the Option C fallback decision above.
- `has_kalshi_match=False` is kept as NaN, not 0 — "no market existed to check" differs from "a market existed but was quiet."

Flags: `no_settlement_seen = tradingStatus` never reached 3 (settled) in our window; `thin_trading(<5 updates)` = fewer than 5 distinct odds states seen; `no_manifest_entry` = match ID present in live data but not in the manifest. All three are capture-window artifacts, not corruption.

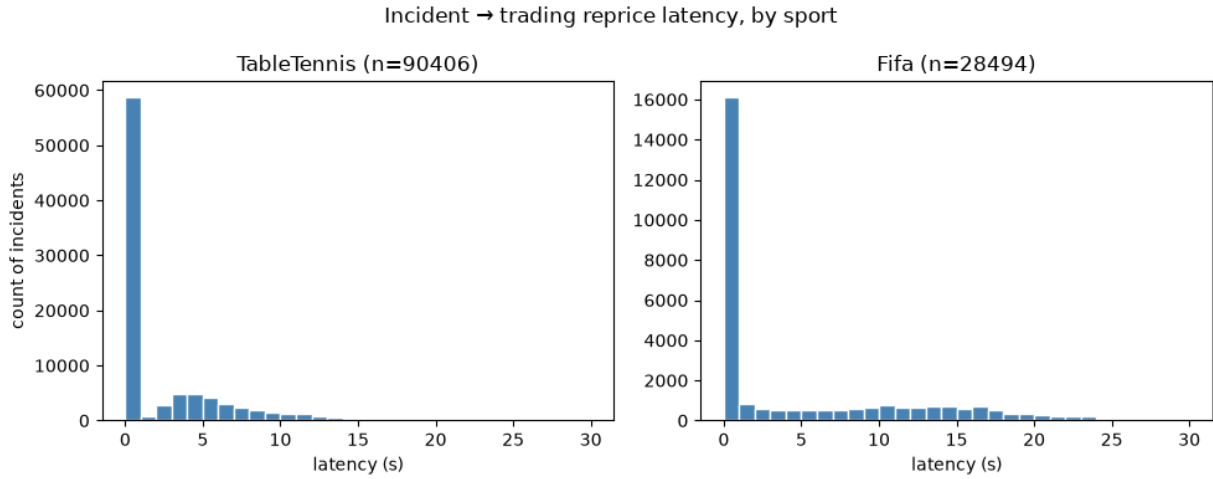


The tall bar at `num_odds_updates=1` is **capture-boundary truncation**: those matches' start times cluster in the final hours of the 2-day capture, so they got one pre-match snapshot before the capture ended and no game ever kicked off. Confirmed: this group reaches settlement only 2–4% of the time, vs. 99% for matches with real trading history.

Part 2(a), Item 1: Incident→Trading Latency

Method. Part 0's cross-correlation needs two *signed* series; an incident has no sign. Instead we use an **event-study response time**: for every non-administrative incident, find the first later `trading` update whose probability actually differs, and record the elapsed time (30s cap; undetected incidents kept as `detected=False`, not dropped). Match-winner markets: TT uses `(marketType,resultType,interval)=(1,6,1)` (clean 2-outcome moneyline); FIFA uses `(6,1,1)`, a Draw-No-Bet-style proxy (empty `marketValue`, near-symmetric win split with a small push rate) since FIFA has no single obvious moneyline triple (appendix B).

Sport	Detection rate	Median latency	Mean latency	Std
TT	86.2% (90,406/104,869)	0.27s	2.52s	3.95s
Fifa	94.1% (28,494/30,279)	0.18s	5.04s	6.97s



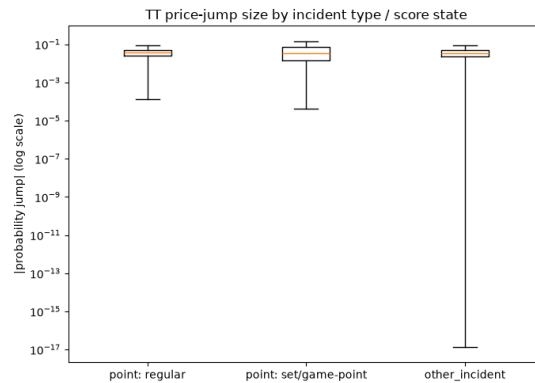
Uncertainty (bootstrap over matches, not incidents — incidents within a match are correlated; 2,000 draws): TT median 0.27s, 95% CI [0.26,0.27]; FIFA median 0.18s, CI [0.17,0.18]; TT–FIFA gap 0.09s, CI [0.08,0.10] — excludes zero.

TT reprices slower than FIFA at the median, consistent with the human-scout-vs-machine-fed hypothesis, though the effect is small in absolute terms. Caveats: (1) sub-second medians for both sports likely reflect near-simultaneous internal backend events, not a human/algorithmic "reaction time" in the intuitive sense; (2) mean $\gg$ median (heavy right tail) means the median is the representative "typical" latency, not the mean.

Part 2(a), Item 2: TT Incident Types, Score State, Price-Jump Size, Calibration

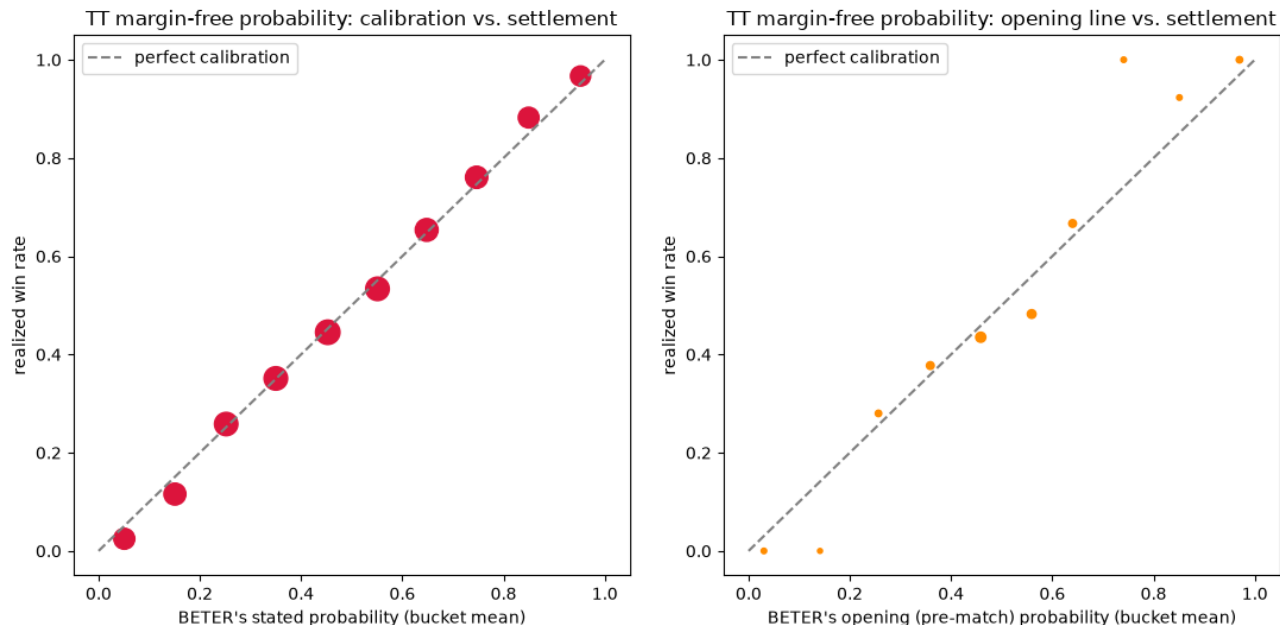
Score-state decoding. `incident.params` key "4" is the current-set point score (resets each set), key "5" is the set score — undocumented, read directly off the data. "A point was scored" = either (a) a within-set +1 change to key "4," or (b) a set-clinching change to key "5" itself (which also resets key "4" — an earlier version missed case (b), mis-bucketing every set-ending point as "other," caught because "other"'s jump size looked suspiciously similar to a regular point's). Set/game-point: standard TT rule (score ≥ 10 , lead ≥ 1); a set-clinching point is automatically one.

Bucket	n	Mean	Median	Std
point: regular	26,548	0.042	0.035	0.029
point: set/game-point	3,871	0.046	0.034	0.043
other_incident	59,594	0.042	0.035	0.031



Median jump size is nearly identical across buckets (≈ 0.034 – 0.035), but set/game-points have a much larger *spread* (std 0.043 vs. 0.029) — not a bigger typical jump: winning an early set barely moves match-win probability, while

winning the deciding set swings it hard toward 0 or 1, and our flag mixes both (a scope choice, not an oversight). **other_incident** moving the price by about the same amount as an ordinary point is a real, mildly surprising finding. **Calibration.** Settlement outcome is known for 442 of 755 manifest TT matches (rest are the capture-boundary-truncated matches from Part 1). Two versions: pooling *every* in-play probability tick (left) vs. only each match’s pre-match *opening* tick (right, one observation/match, verified genuinely pre-match for 440/442 matches — median ~4.2h before that match’s first point; full decile table in appendix F).



Both sit close to the diagonal. In-play shows a small, consistent pattern at the extremes (0.1–0.2 bucket: stated 15.1%, realized 11.6%; 0.9–1.0: stated 95.2%, realized 96.7%) but pools ~38k correlated ticks from the same ~440 matches — late in a lopsided match the scoreboard alone predicts the outcome, so this partly tests whether the scoreboard predicts the scoreboard, not model skill. Opening-line avoids that (one observation/match) but is noisier per bucket ($n=8-154$) and its well-populated middle buckets (0.3–0.7) still land within a few points of the diagonal. **A Brier-score comparison is easy to misread:** opening-line Brier is 0.204 vs. 0.175 in-play — the lower in-play number is *not* evidence of better calibration, just a byproduct of sampling many near-certain probabilities from already-decided games. Net: BETER’s probability is reasonably calibrated on this capture, both pre-match and in-play.

Appendix

A. Why Part 0’s calibration correlations (0.28/0.42/0.61) are low by design. The synthetic generator injects substantial independent per-poll noise into the Kalshi side on top of the real lagged signal, and the signal itself is small relative to Kalshi’s 1-cent quantization — deliberate stress-testing under noise, not a real-data benchmark. The estimator still recovers the correct lag despite the low peak correlation, which is the property that matters.

B. FIFA match-winner market identification. FIFA has dozens of (marketType,resultType, interval) triples; most are handicap/totals markets (numeric marketValue, ~1–2% “push” outcome). Triple (6,1,1) has an empty marketValue and a near-symmetric win split (462 vs. 458 of ~927 settled, plus ~0.8% push) — consistent with Draw-No-Bet: voids on a real-world draw, otherwise pays the winner. Used with this caveat; not a pure moneyline like TT’s (1,6,1).

C. Kalshi crosswalk. Tickers encode KX<SERIES>-<DATE><HHMM><TAG1><TAG2>-<SIDE>. Matching requires (i) correcting the embedded timestamp from America/New_York to UTC and (ii) scoring candidate team-tag matches against name-derived abbreviations, accepting only a strong combined score *and* a corrected start time within 1 hour. Found exactly 2 matches (both Counter-Strike) of 1,465 — the basis for concluding TT/FIFA had no live Kalshi counterpart.

D. Bootstrap methodology. Resampling is over *matches*, not incidents (incidents within a match are correlated). Each of 2,000 draws resamples matches with replacement within a sport, pools every detected incident from the resampled matches, and recomputes the median latency; percentiles across draws give the 95% CIs.

E. Deliverable-table flag glossary. no_settlement_seen: status never reached 3 before capture end. thin_trading(<5 updates): fewer than 5 distinct odds states observed. no_manifest_entry: match ID in live data but not the manifest. Flags can co-occur (semicolon-joined).

F. Opening-line calibration, full decile table (440 matches; 2 dropped where the first tick wasn’t verified pre-match):

Bucket	n	Mean p.	Win rate	Bucket	n	Mean p.	Win rate
0.0–0.1	13	0.030	0.000	0.5–0.6	85	0.559	0.482
0.1–0.2	8	0.141	0.000	0.6–0.7	51	0.640	0.667
0.2–0.3	25	0.257	0.280	0.7–0.8	13	0.741	1.000
0.3–0.4	53	0.359	0.377	0.8–0.9	13	0.851	0.923
0.4–0.5	154	0.459	0.435	0.9–1.0	25	0.970	1.000

The two lowest buckets' 0% realized win rate is consistent with sampling noise at that n (expected well under 1–2 wins), not miscalibration.

Part 3 (edge backtest, fallback variant (b)) and the final ≤ 4 -page write-up are in progress.